

BTEC Assignment Brief

Qualification	Pearson BTEC Level 5 Higher Nationals in Digital Technologies
Unit number and title	Unit 19: Business Intelligence
Learning aim(s)	By the end of this unit students will be able to: LO1 Discuss business processes and the mechanisms used to support business decision-making; LO2 Compare the tools and technologies associated with business intelligence functionality; LO3 Demonstrate the use of business intelligence tools and technologies LO4 Discuss the impact of business intelligence tools and technologies for effective decision-making purposes and the legal/regulatory context in which they are used;
Assignment title	Business Intelligence (BI)
Assessor	Ulug`bek Tursunaliev
Issue date	April 16, 2025
Hand in deadline	

Design and development of a business intelligence (BI) dashboard project

Objective

The objective of this assignment is to develop a business intelligence (BI) dashboard that supports decision-making for a chosen business field. The dashboard should visualize key performance indicators (KPIs) and provide actionable insights based on data analysis. This task will assess your ability to understand business processes, compare BI tools and technologies, demonstrate the use of BI tools, and evaluate the impact of BI on organizational decision-making while considering legal and regulatory contexts.

Tasks:

1. Business Analysis

- **Purpose:** To understand business processes and decision-making mechanisms within a business field or an organization (**LO1**).
- **Tasks:**
 - Select a real or hypothetical organization / business field (e.g., a retail company, healthcare provider, educational institution, logistics and supply chain, etc).
 - Examine its business processes, focusing on those related to decision-making, such as management (e.g., forecasting), inventory management, customer segmentation, operational (e.g., sales, purchasing), supply chain optimization or support processes (e.g., accounting, technical support).
 - Identify the types of data used in these processes, such as *structured* (e.g., databases), *unstructured* (e.g., emails), and *semi-structured* (e.g., XML files) data.
 - Evaluate the current information systems and technologies used or needed at operational, tactical, and strategic levels to support decision-making (e.g., transaction processing systems, management information systems, decision support systems).

2. Tools and Technologies

- **Purpose:** To compare and select appropriate BI tools and technologies (**LO2**).
- **Tasks:**
 - Compare at least two BI tools or technologies, such as Tableau, Power BI, QlikView, open-source alternatives like Apache Superset, self-developed BI visualization tools or other web-based alternatives.
 - Justify your selection of a specific BI tool for this project, explaining its key features (e.g., data visualization capabilities, ease of use, integration options) and why it suits your organization's or business processes needs.

	3. Design and Implementation ● Purpose: To demonstrate the use of BI tools by creating a functional dashboard (LO3). ● Tasks: ○ Design a BI dashboard that visualizes key metrics and KPIs for your chosen organization or business process (e.g., sales trends, customer satisfaction scores, inventory levels). ○ Implement the dashboard using your selected choice of BI tool, ensuring it includes appropriate visualizations (e.g., charts, graphs, histograms) and interactivity (e.g., filters, drill-down options). ○ Ensure the dashboard is user-friendly and functional, with a clear and intuitive interface. ○ Critically review your design, explaining how it meets the user or business requirements (e.g., how it addresses specific decision-making needs). 4. Testing and Improvement ● Purpose: To refine the BI dashboard based on feedback (LO3). ● Tasks: ○ Gather feedback from potential users, peers, or stakeholders on your dashboard (e.g., through a survey, interview, or usability testing). ○ Refine and improve the dashboard based on the feedback received, documenting the changes made (e.g., adjusting visualizations, adding new features, improving navigation). 5. Impact and Legal Considerations ● Purpose: To evaluate the impact of BI for effective decision-making purposes and consider legal/regulatory issues (LO4). ● Tasks: ○ Discuss how your BI dashboard can contribute to effective decision-making in the organization or business process (e.g., by improving efficiency, providing faster results, enhancing customer service, or supporting strategic planning). ○ Research and provide examples of organizations that have successfully used BI tools to enhance or improve operations (e.g., how companies like Amazon or Walmart use BI for supply chain optimization). ○ Explore the legal and regulatory issues related to data usage in BI, such as data privacy laws (e.g., GDPR), data security, and ethical considerations (e.g., ensuring unbiased data analysis).
Assignment Report	Submission Requirements: 1. Report: Submit a written report that includes:

Dashboard Sources of information to support you with this Assignment	• An introduction to the organization or a business field and its business processes. • Analysis of current information systems and technologies. • Justification for the selected BI tool. • Design documentation for the dashboard (e.g., wireframes, data sources, KPIs). • Screenshots or a live demonstration of the dashboard. • Feedback and improvement log, detailing user feedback and changes made. • The report should be well-structured, with clear headings, and include references to any sources used (e.g., textbooks, websites, journals). 2. BI Dashboard: Provide access to the BI dashboard if possible (e.g., a shared link to a cloud-based dashboard) or include detailed screenshots showing its functionality. Important: • Ensure the report is concise yet comprehensive, typically 3,000–5,000 words (excluding <i>cover page, figures, graphs, reference/bibliography list</i> and <i>appendices</i>), depending on institutional guidelines. • Use a professional format, with proper citations in a recognized style (e.g., <i>APA style, MLA style or Harvard style</i>). Books: 1. “Business Intelligence Guidebook: From Data Integration to Analytics” Author: Rick Sherman 2. “Business Intelligence Roadmap: The Complete Project Lifecycle for Decision-Support Applications” Authors: Larissa T. Moss and Shaku Atr 3. “Business Intelligence in Plain Language: A practical guide to Data Mining and Business Analytics” Author: Jeremy Kolb 4. “Successful Business Intelligence: Unlock the Value of BI & Big Data” Author: Cindi Howson Websites: 1. https://www.tableau.com/ 2. https://www.qlik.com/us 3. https://www.zoho.com/analytics/

Learning Outcomes and Assessment Criteria

Pass	Merit	Distinction
LO1 Discuss business processes and the mechanisms used to support business decision-making		LO1 and LO2 D1 Evaluate a range of information systems and technologies that can be used to support organisations at operational, tactical and strategic levels.
P1 Examine, using examples, the terms 'business processes' and 'supporting processes'.	M1 Differentiate between unstructured and semi-structured data within an organisation.	
LO2 Compare the tools and technologies associated with business intelligence functionality		
P2 Compare the types of support available for business decision-making at varying levels within an organisation. P3 Determine, with examples, what business intelligence is and the tools and techniques associated with it.	M2 Justify, with specific examples, the key features of business intelligence functionality.	
LO3 Demonstrate the use of business intelligence tools and technologies.		D2 Critically review the design in terms of how it meets a specific user or business requirement and identify the customisation that has been integrated into the design.
P4 Design and implement a business intelligence tool, application or interface that can perform a specific task to support problem-solving or decision-making at an advanced level. P5 Improve the business intelligence tool by gathering feedback and refining it.	M3 Customise the solution to ensure that it is user-friendly and has a functional interface.	
LO4 Discuss the impact of business intelligence tools and technologies for effective decision-making purposes and the legal/regulatory context in which they are used.		
P6 Discuss how business intelligence tools can contribute to effective decision-making. P7 Explore the legal issues involved in the secure exploitation of business intelligence tools.	M4 Research with specific examples organisations that have used business intelligence tools to enhance or improve operations.	D3 Evaluate how organisations could use business intelligence to extend their target audience and make them more competitive within the market, taking security legislation into consideration.